Module 6 Capstone Video Script

Target duration: 8 minutes 30 seconds | Maximum 10 minutes | Maximum 250 MB

Public app: https://group19-network-intrusion-detection.streamlit.app/
Public repository: https://github.com/adugnaNecho/Group19-network-intrusion-detection

Use this as a guide, not as a claim that every activity has already been performed. Adapt it to what the group actually demonstrates. Keep the screen recording factual and show the current public app.

0:00-0:40 | Introduction and problem

Introduce Group 19 and the five-class connection-record classification problem. Explain that the prototype classifies uploaded NSL-KDD-style records; it is not a live packet sensor or traffic-blocking system.

0:40-1:40 | Dataset and split

Show public NSL-KDD KDDTrain+ and KDDTest+, 41 input features, five-class mapping and the stratified validation split. Explain that KDDTest+ is reserved for final evaluation.

1:40-2:40 | Model pipeline

Explain numeric/categorical preprocessing, SMOTENC within the training pipeline and Random Forest. State that synthetic minority examples are training-only; validation and KDDTest+ are not oversampled.

2:40-4:20 | Live prototype

Open the public Streamlit app. Show the disclaimer, sample CSV, 41-feature input upload, Predict intrusion class, output class, maximum probability, threat severity, class probabilities and downloadable results. Describe the supplied DoS/1.00/High screenshot as an illustrative interface output, not verified ground truth.

4:20-5:40 | Dashboard and evaluation

Show grouped validation/test bars, class-wise table, confusion matrix and FPR panel. Report validation accuracy 99.89%, Macro F1 98.26%, and KDDTest+ accuracy 74.62%, Macro F1 51.51%. Report recall by class and call out U2R 8.96% and R2L 7.20%.

5:40-6:30 | Testing and deployment

Show the public GitHub repository and Streamlit application. Explain GitHub source/artifact management and Streamlit hosting. State that 29 automated tests passed in finalization QA; distinguish software tests from model performance. FastAPI and Docker are optional included paths, not separate public production services.

6:30-7:40 | Monitoring and maintenance

Discuss the proposed plan for latency/error monitoring, verified labels, drift screening, reviewed retraining, artifact version/checksum records, privacy controls, human oversight, shadow testing and rollback. Clearly identify these as proposed, not active continuous services.

7:40-8:30 | Reflection and close

Summarize achievements, technical challenges, generalization gap and rare-class limitations. Describe future work on newer authorized datasets, rare-class recall, calibration and controlled pilot evaluation with analyst oversight.

Closing statement

“Group 19 has progressed from model development to a publicly accessible research prototype. The application demonstrates CSV-based five-class prediction and an analytics dashboard. The independent KDDTest+ results show that generalization, particularly for U2R and R2L, needs improvement. We therefore present this work as a functional prototype with a documented monitoring and maintenance plan, not as a production intrusion prevention system.”

Recording reminders

Stay under 10 minutes and 250 MB. Show the public app and actual input/output; separate validation from independent-test results; disclose benchmark and SMOTENC synthetic training data; confirm Cloud hotfix commit 8941226 before showing the FPR panel; include each member's actual contribution; and hide credentials or private network data.